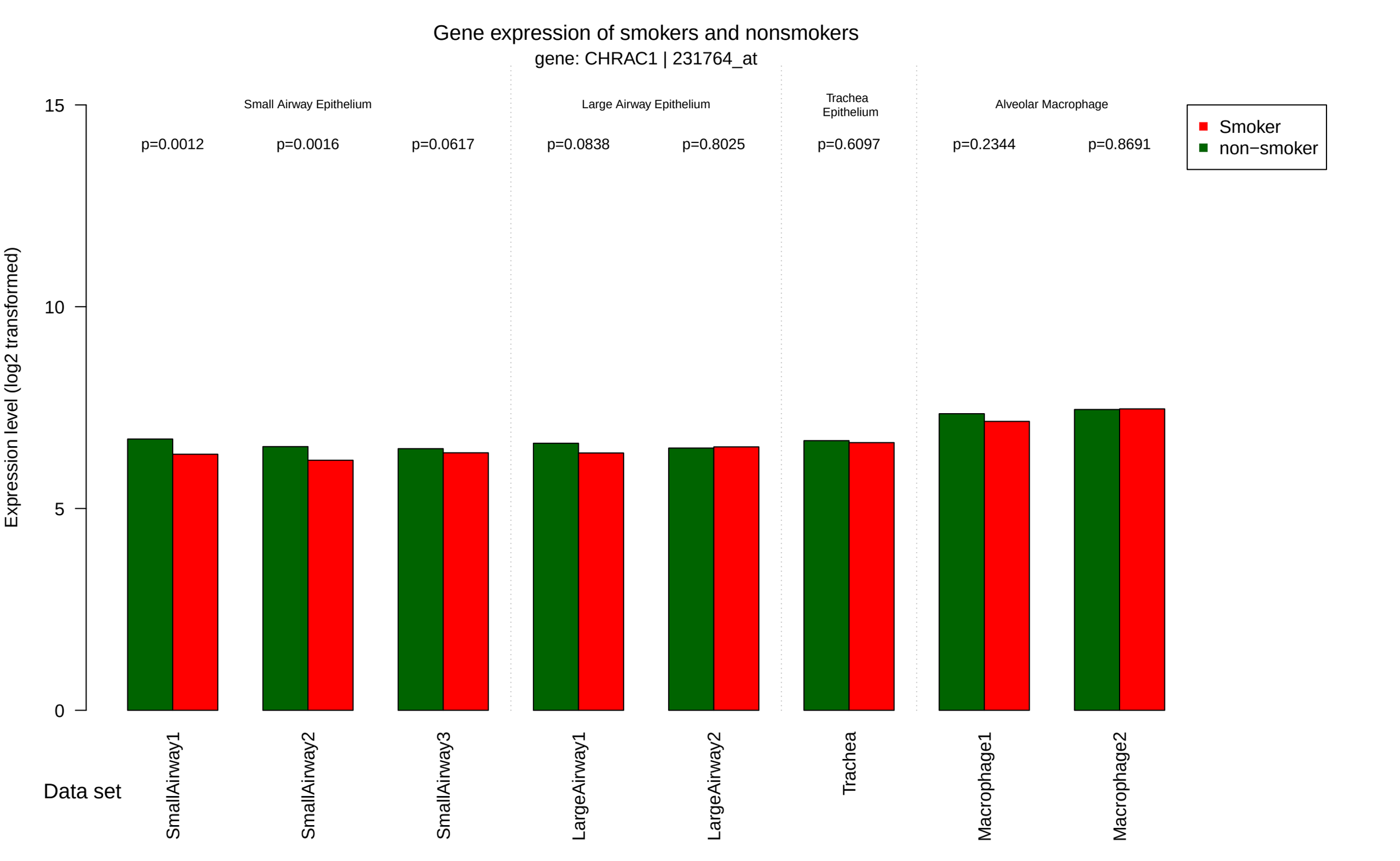
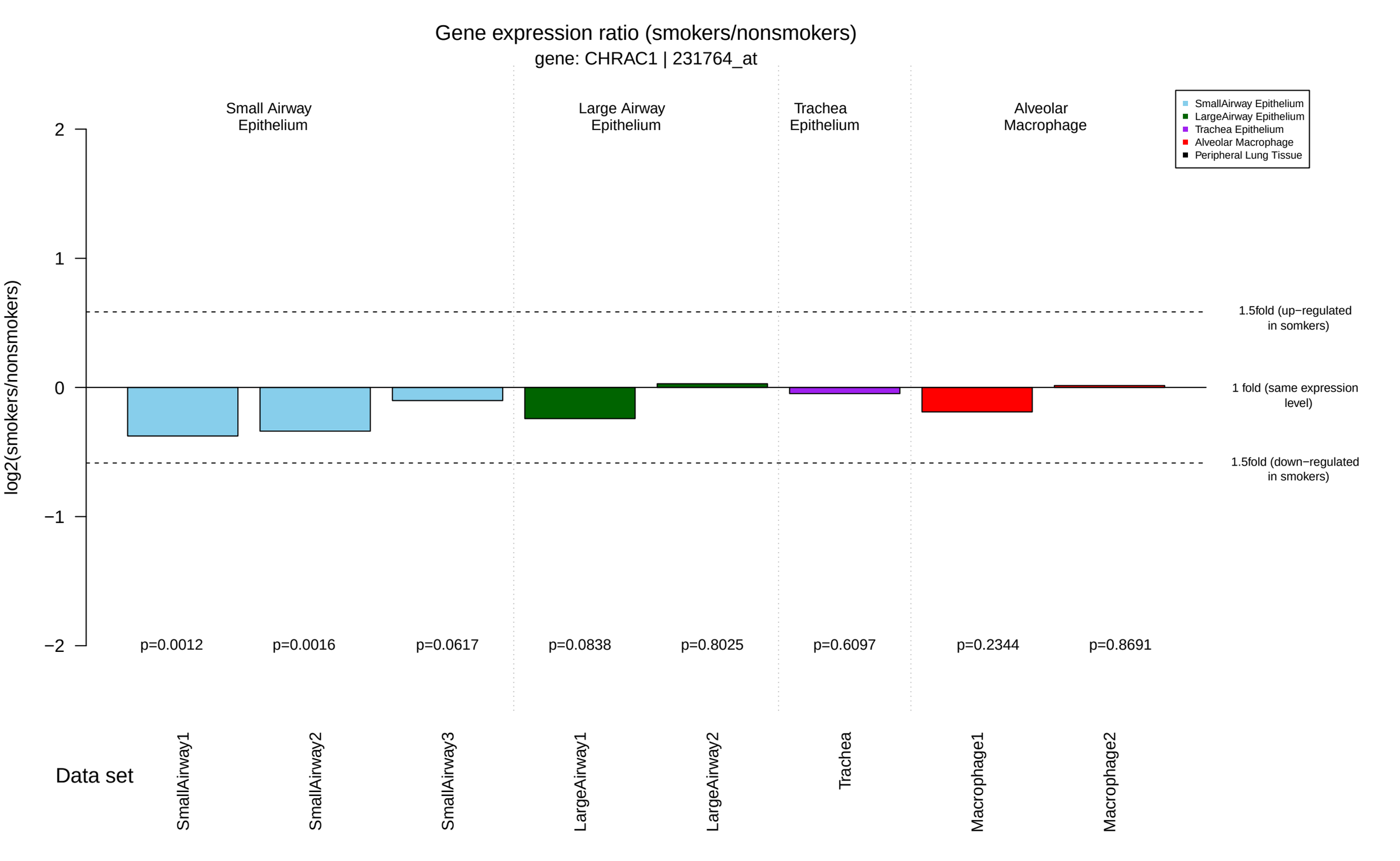
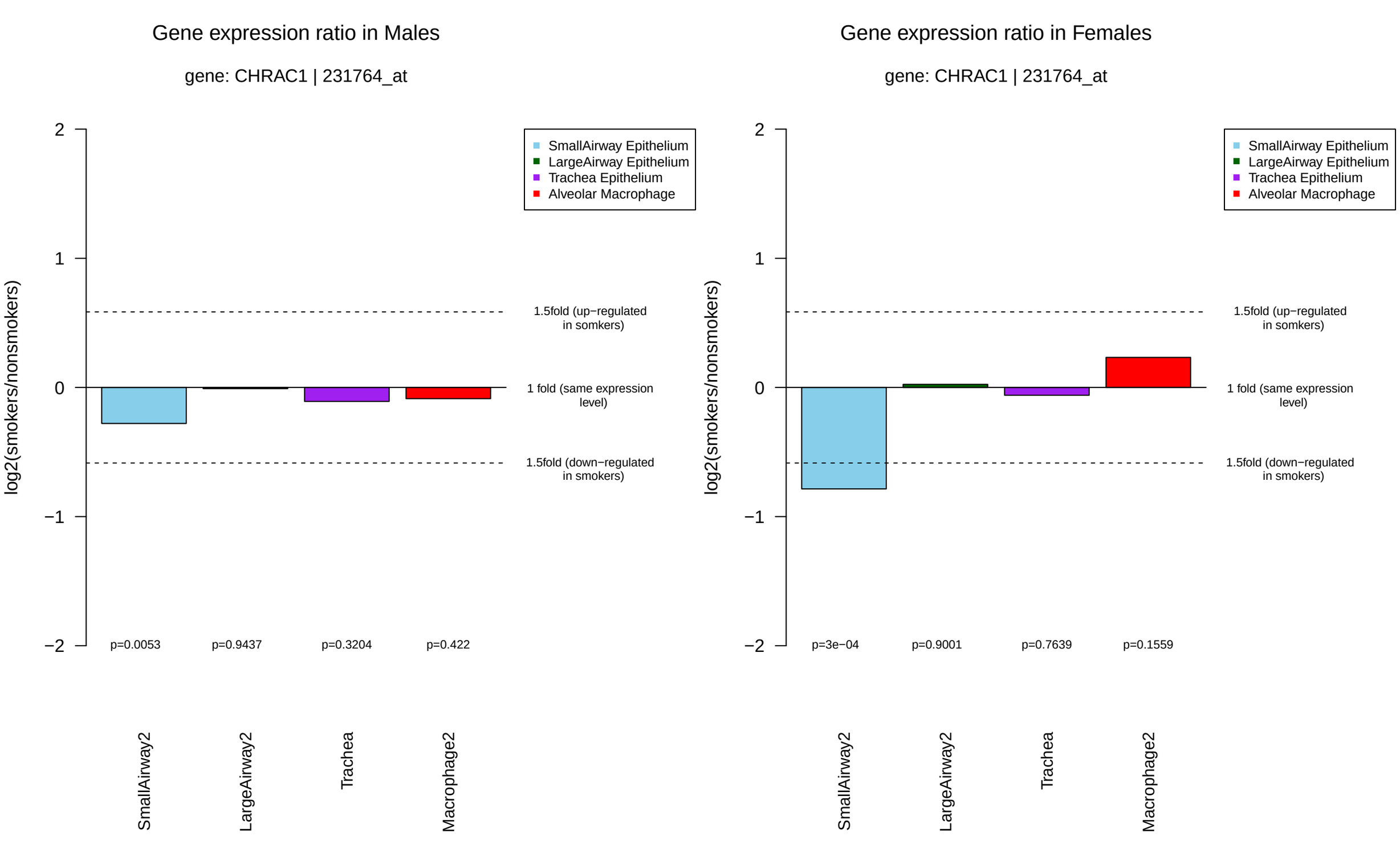
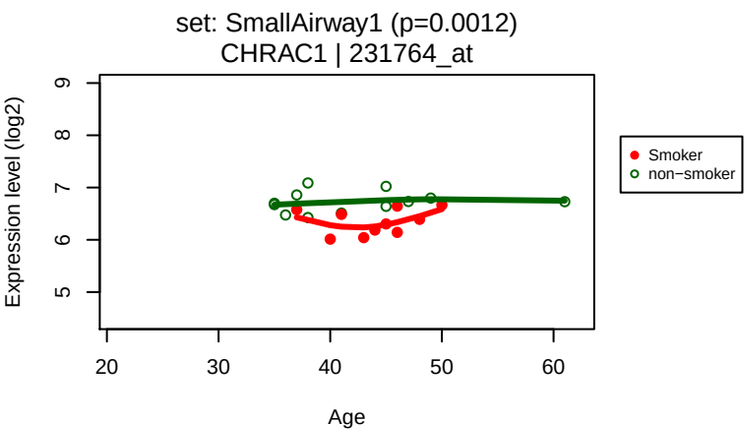
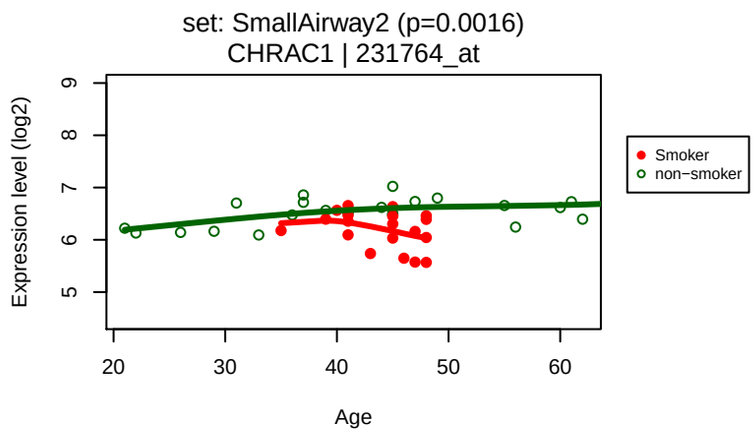
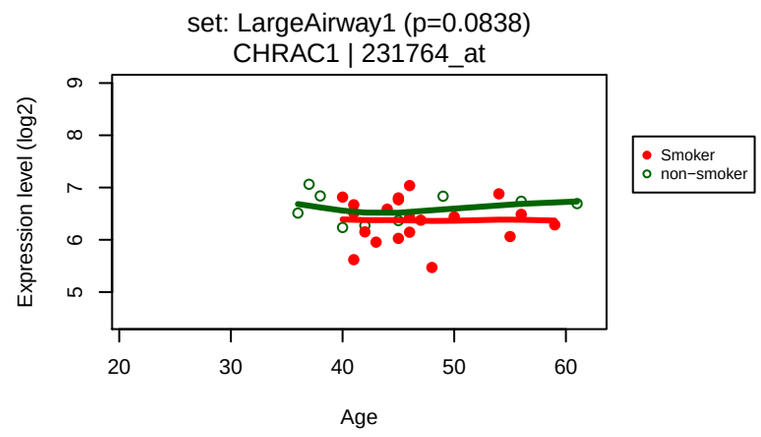

Gene expression of smokers and nonsmokers
gene: CHRAC1 | 231764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0012
p=0.0016
p=0.0617
p=0.0838
p=0.8025
p=0.6097
p=0.2344
p=0.8691
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

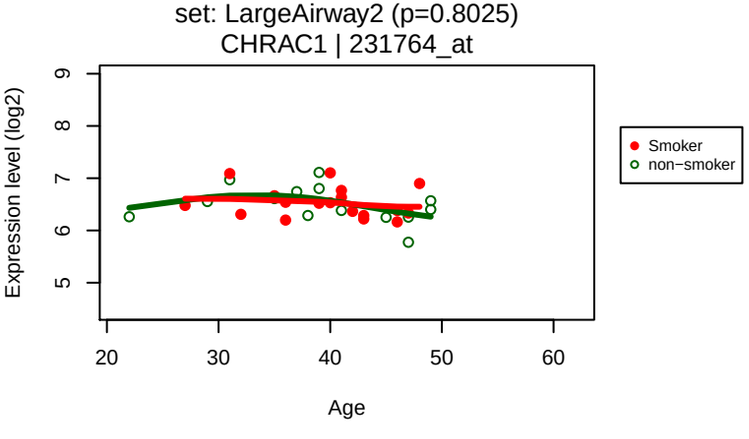

Gene expression ratio (smokers/nonsmokers)
gene: CHRAC1 | 231764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0012
p=0.0016
p=0.0617
p=0.0838
p=0.8025
p=0.6097
p=0.2344
p=0.8691
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

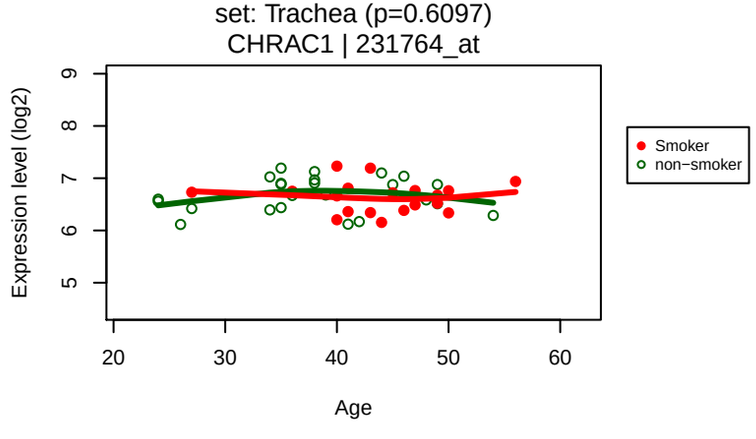

Gene expression ratio in Males
gene: CHRAC1 | 231764_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0053
p=0.9437
p=0.3204
p=0.422
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRAC1 | 231764_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=3e−04
p=0.9001
p=0.7639
p=0.1559
SmallAirway2
LargeAirway2
Trachea
Macrophage2

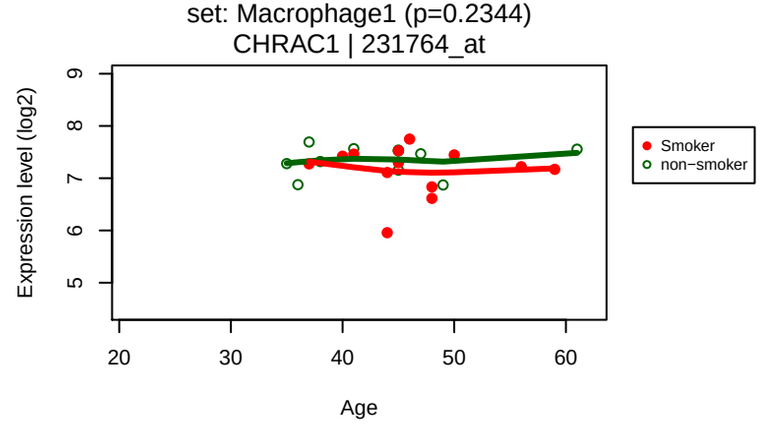

set: SmallAirway1 (p=0.0012)
CHRAC1 | 231764_at
Expression level (log2)
Age
Smoker
non−smoker

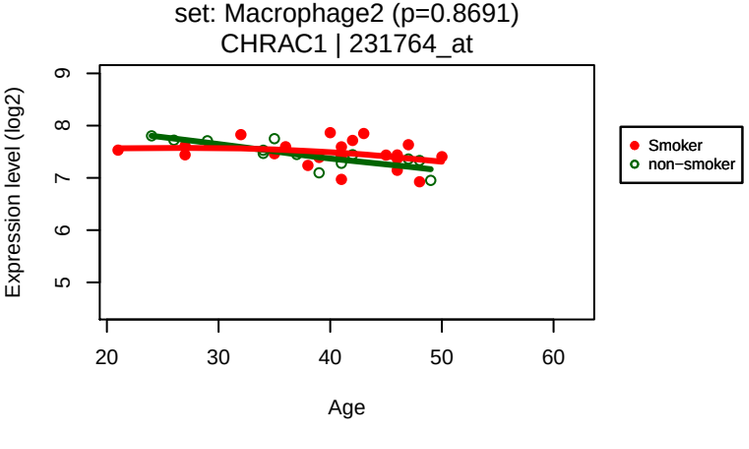

set: SmallAirway2 (p=0.0016)
CHRAC1 | 231764_at
Expression level (log2)
Age
Smoker
non−smoker

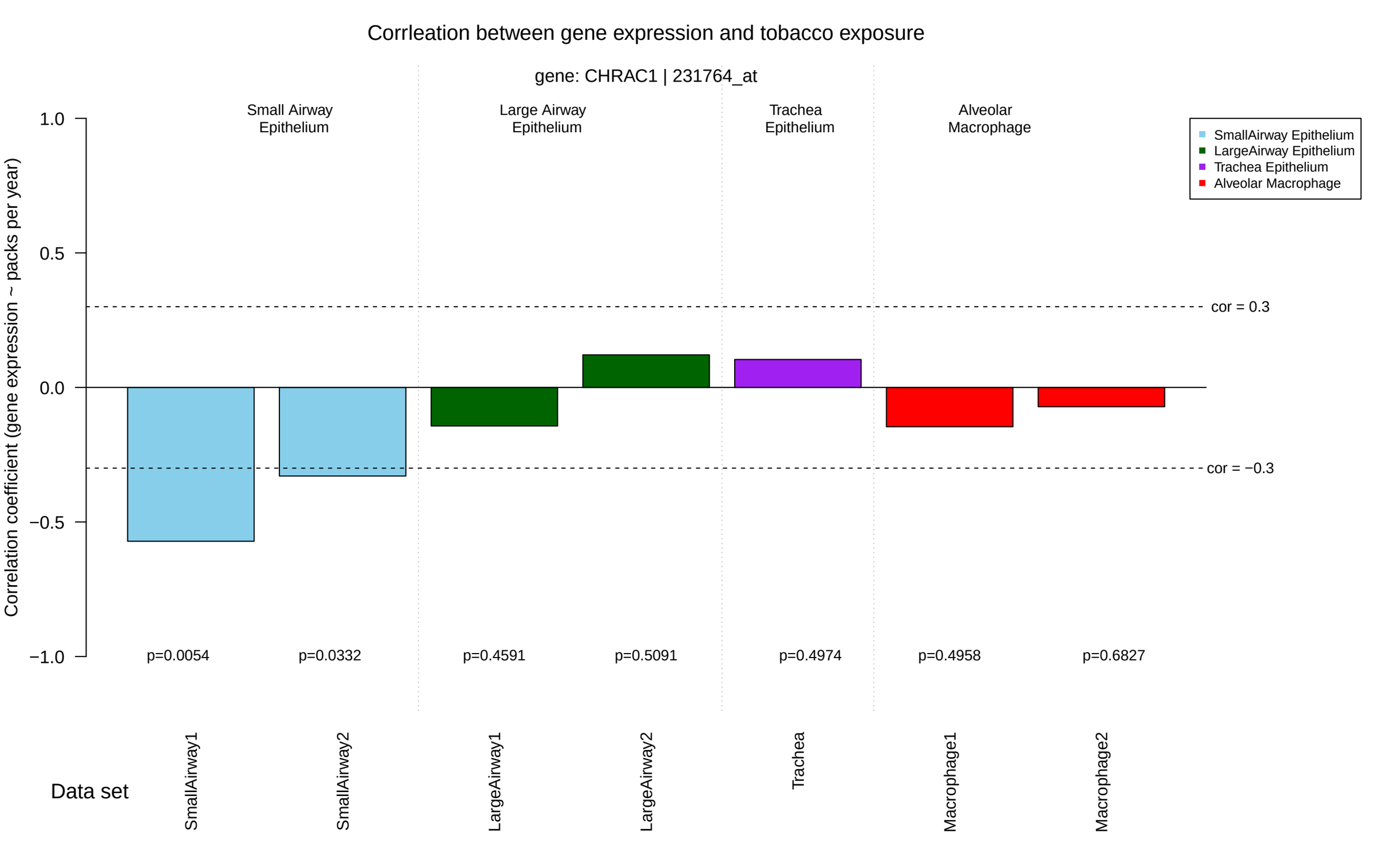

set: LargeAirway1 (p=0.0838)
CHRAC1 | 231764_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8025)
CHRAC1 | 231764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6097)
CHRAC1 | 231764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2344)
CHRAC1 | 231764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8691)
CHRAC1 | 231764_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRAC1 | 231764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0054
p=0.0332
p=0.4591
p=0.5091
p=0.4974
p=0.4958
p=0.6827
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2